

High-Frequency Market Making Infrastructure Six: Data Scope for Market Quotes, Part Two

SK (@StepOneAi) · 7 August 2026 · posted on X · translated from Chinese

The previous article discussed the outer-layer average for funding rates. This article discusses the thing being averaged itself.

The premium index is not calculated using the mid-price or best price.

$$P = [\max(0, \text{Impact Bid} - \text{index price}) - \max(0, \text{index price} - \text{Impact Ask})] / \text{index price}$$

The definition of Impact Bid / Impact Ask is: the average execution price after consuming a fixed nominal amount (Impact Margin Notional, IMN).

So it measures the deviation at a certain depth, not the deviation in the order book. The original design intent is very clear: to prevent a thin single level of hanging orders from influencing the rate. But the problem that comes with it is: what exactly is the value of IMN.

I've seen two ways of setting it. One is a configuration value: IMN is a hardcoded USDT number written in the contract specifications, unrelated to other parameters, and announced separately when changed. The other is a derived value: IMN is back-calculated from the initial margin rate of the contract's highest leverage tier as $\text{IMN} = \text{a fixed margin amount} / \text{initial margin rate of the highest leverage tier}$.

In the latter approach, the initial margin rate in this formula is the denominator: the higher the maximum leverage, the lower the initial margin rate, the larger the nominal amount leveraged by the same margin amount, and thus the larger the IMN. As a result, on platforms using the derived method, the rates for the same internal entity but different contracts are measured at different depths. Contracts with higher maximum leverage have larger IMN and deeper sampling. Those with lower maximum leverage have shallower sampling. That early confusion in our horizontal comparison—where order book shapes looked the same, but sampling points weren't aligned, so the numbers naturally didn't match—makes sense now.

However, the "depth" here refers to absolute nominal amounts. High-leverage contracts are usually also the ones with the best liquidity, so the same amount placed into a mainstream order book might only consume a few levels to suffice. Placed into a thin order book, it might already penetrate deep inside. Relative to order book thickness, the direction might even reverse. To judge exactly how deep this indicator is looking, you have to compare IMN with the actual order book thickness of that contract; looking at IMN alone isn't enough.

At the same time, this depth changes. Leverage and margin tiers are adjustable parameters; one adjustment, and IMN takes on a new value. The market quotes haven't changed, the order book hasn't changed, but the measurement position has. And even if IMN hasn't budged a single unit, if the order book thickness changes, that relative depth still shifts.

This type of tier adjustment often concentrates in phases of amplified volatility and rising open interest—precisely when the rates themselves are under the most scrutiny and the order book is the least stable. Two things happening at once. This is an observation, not a rule, but it's enough to put it on the list of event sources needing monitoring.

As for which type your platform falls into, you'll have to check the contract specifications and rate descriptions line by line to confirm. Under these two setups, the answer to whether IMN is a constant is the opposite, and that answer determines whether you need to build a parameter history for it.

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Archived text of the X post (English as rendered by X's translation of the Chinese original).